Roll No. | | | | | | | |

Even End Semester Examination 2025

Name of the Program: B. Tech
Name of the Course: Basic Electrical Engineering

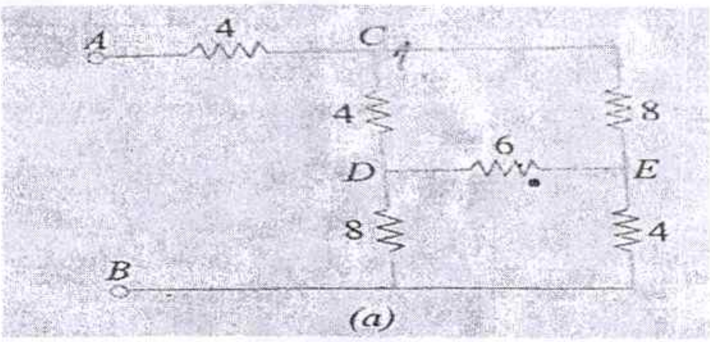
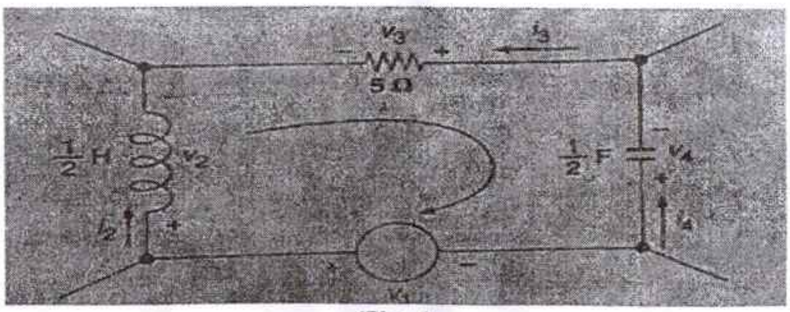
Semester: II
Course Code: TEE 201

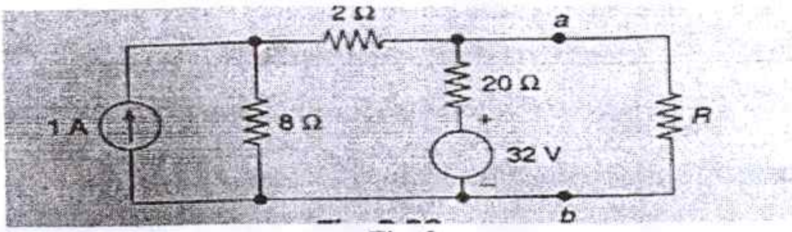
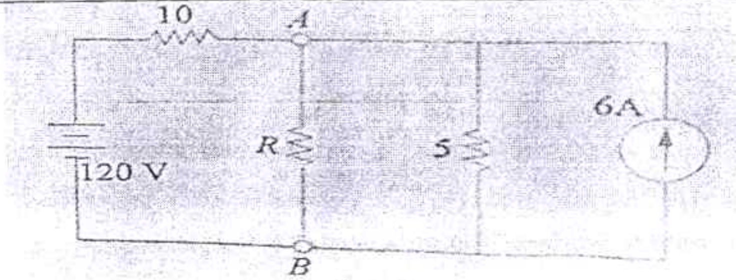
Time: 3 Hrs

Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	Define along with examples (i) Unilateral and Bilateral element (ii) Active and Passive elements (iii) Discuss the basic characteristics of resistance, capacitance and inductance. (iv) Define form factor and peak factor for an AC circuit.	**CO1**
(b)	Write the formula for star to delta and delta to star conversion using suitable diagram. Find the equivalent resistance seen between the terminals A-B of Fig. 1.  **Fig. 1**	
(c)	Consider the mesh of Fig. 2 which forms a part of larger circuit. Given $v_1 = 10 \sin t$, $i_2 = 4 \sin t$, $i_3 = 2 \cos t$. Determine i_4.  **Fig. 2**	
Q2	**(20marks)**	
(a)	State Norton's Theorem. Explain the procedure to find Norton's equivalent circuit with a suitable example.	**CO2**

(b)	Find the Thevenin equivalents of the circuit of Fig. 3 as seen at terminal ab.  **Fig. 3**	
(c)	For a full wave rectified alternating current output, obtain the RMS value, average value, form factor, and peak factor.	
Q3	**(20marks)**	CO3
(a)	Perform the following indicated operations and obtain results in polar form: (i) (60+j80) + (30-j40) (ii) (12-j6) – (40-j20)	
(b)	A coil of resistance 8 ohm and inductance 0.1 H is connected in series with a condenser of capacitance 160 μF across 230 V, 50 Hz supply. Calculate: (i) inductive reactance (ii) Capacitive reactance (iii) Circuit impedance (iv) Current (v) Power factor	
(c)	Draw the waveforms and phasor diagram for single phase alternating current and voltage for series R-L circuit and hence derive the expression for average power.	
Q4	**(20marks)**	CO4
(a)	Three voltages represented by: V_1=20 sin ωt V_2=30 sin (ωt-π/4) V_3=40 cos (ωt+π/6) Act together in a circuit. By using phasor diagram find an expression for the resultant voltage	
(b)	Differentiate between electrical wire and cable. Give classification of electrical cable. Explain any two classifications in detail.	
(c)	For a single phase series R-C ac circuit define the following terms with the phasor diagram: (i) voltage triangle (ii) impedance triangle (iii) power triangle	
Q5	**(20 Marsk)**	CO5
(a)	What do you understand by Switchgear? What are the essential features of switchgear? Give classification of switchgear.	
(b)	What do you understand by "Electrical Earthing"? What are the basic needs of electrical earthing? Define its basic components. Also give classification of electrical earthing?	
(c)	Calculate the value of R which will absorb maximum power from the circuit of Fig 4. Also compute the value of maximum power.	
	 Fig.4	